

# Construction-Site PPE Safety Detection

YOLO11 Nano | Reproducible cloud workflow | Evidence-led iteration

---

## MAICEN-0526 - Group 5 – M4U3 Assignment Submission

Submitted by Group 5:

Dru (Nagendrudu Panangipalli)  
Mohamed Mostafa Abdallah Elghandour  
Ahmed Emad  
Ahmed Nashat Solyman Shaheen

22 September 2026

# 96.93%

M1 held-out mAP50

The central finding

**Strong held-out performance did  
not eliminate external domain-  
shift failures.**

# From AECO safety need to a measurable detection task

Seven-class PPE ontology, versioned dataset and explicit success criteria.

AECO problem

Support visual review of construction-site PPE and selected safety conditions across varied site imagery.

Dataset

3,858 annotated images | 7 classes | Roboflow Universe | CC BY 4.0.

Actual split

3,287 train (85%) | 340 validation (9%) | 231 test (6%).

## Class ontology

Person

Helmet

Non-Helmet

Vest

Gloves

Shoes

bare-arms

Important ontology limitation

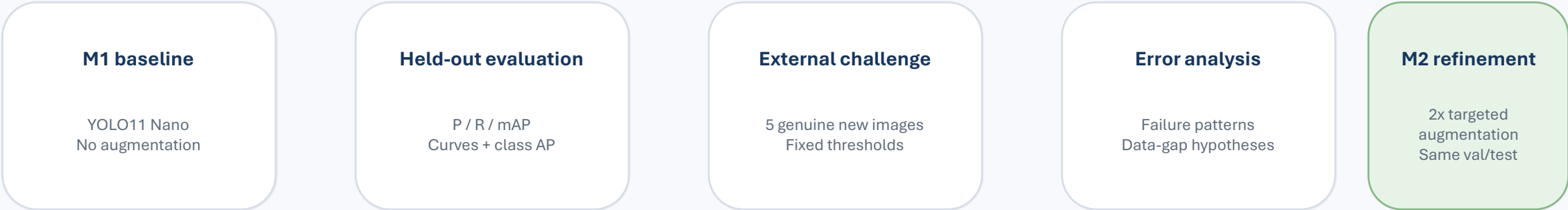
Non-Helmet exists, but there are no Non-Vest, Non-Gloves or Non-Shoes classes. Missing a positive PPE detection is therefore not automatically a safety violation.

Pre-processing: 640 x 640 resize using fit with black-edge padding.

**Success criterion: strong held-out detection plus defensible evidence of generalisation, failure modes and responsible-use limits.**

# A controlled M1 → challenge test → M2 workflow

Only the training augmentation changed; held-out validation and test sets were retained.



**M2 targeted augmentation**

**Horizontal flip • Crop 0-10% • Rotation -15° to +15° • Brightness -15% to +15% • 2x training multiplier**

Experimental discipline

M2 targeted augmentation was informed by the external generalisation weaknesses observed in M1. As M2 was not re-evaluated on the same external challenge set, conclusions are limited to the controlled held-out results, without inferring unverified external performance improvements.

**Rationale for Adaptations to the Reference Model Architecture and Dataset Split**

As an applied research/experiment extension to the reference configuration our Group 5 , YOLO11 Nano was adopted to explore an alternative contemporary lightweight Ultralytics model architecture, while an 85/9/6 train/validation/test dataset split was used to maintain distinct held-out validation and test sets. These configurations were applied consistently across M1 and M2, preserving controlled experimental comparability, and are explicitly documented to ensure transparency and reproducibility. This extension also provided practical insight into applying an alternative YOLO architecture and dataset-splitting strategy to the same AEEO object-detection problem while retaining the core experimental and evaluation framework.

# Strong held-out performance established a credible baseline

The baseline is excellent internally, but these numbers are not equivalent to real-world accuracy.



## Test-set class mAP50



### What the baseline tells us

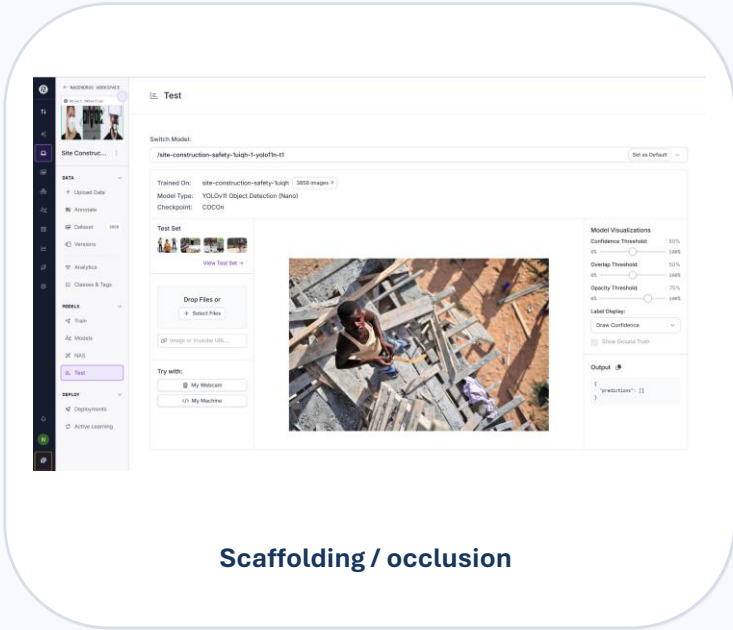
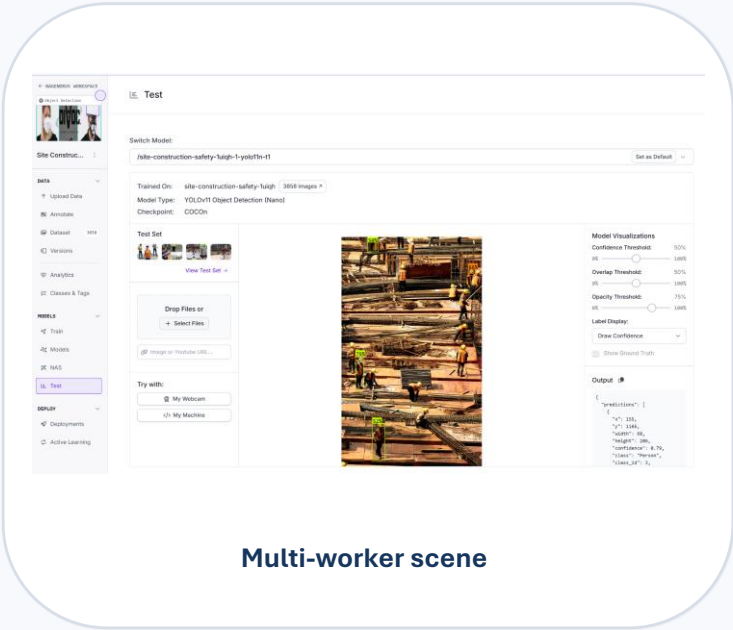
Validation and test mAP50 were both approximately 97%, with no obvious held-out collapse. bare-arms was the weakest test class at 91% mAP50.

### What it does not tell us

Held-out performance does not prove robustness to unfamiliar cameras, framing, lighting, PPE appearance, occlusion or worker scale.

# External images exposed the deployment gap

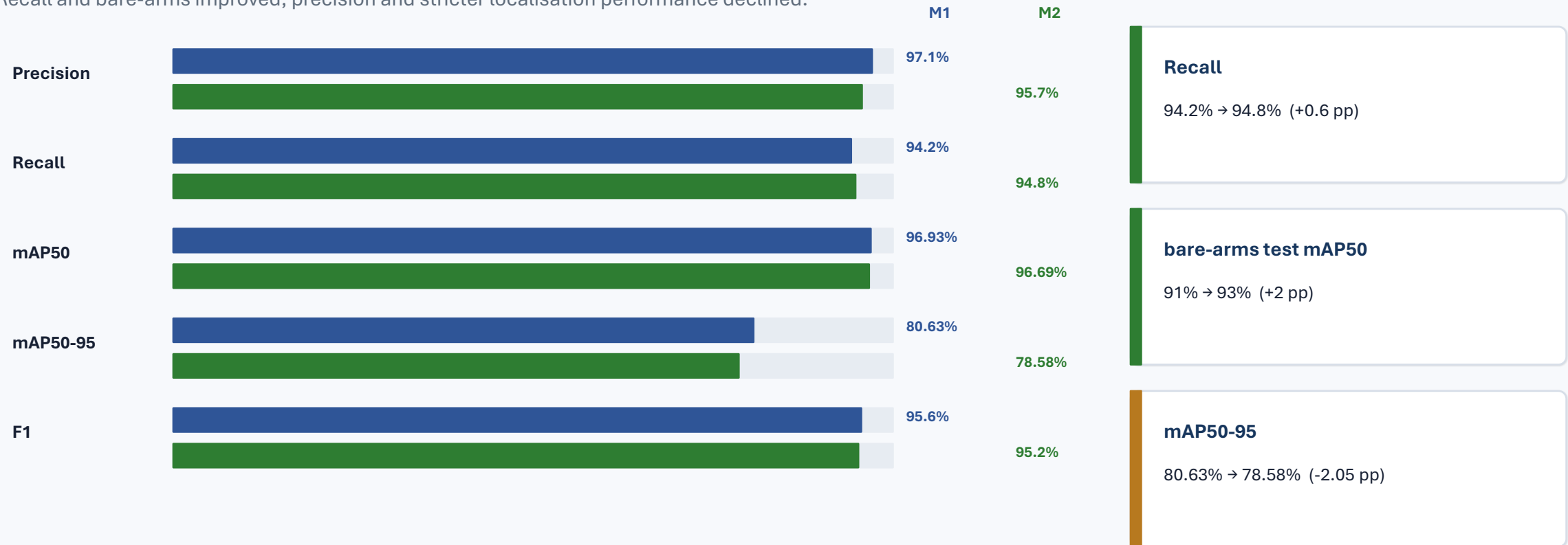
Five genuine challenge images revealed failure modes hidden by the held-out aggregate metrics.



**Observed pattern** Misses involved Non-Helmet, bare-arms, Gloves, partially framed workers, complex scenes and distant/background people. External testing was qualitative - not formal mAP - because these images were not manually annotated as a ground-truth set.

# Targeted augmentation changed the trade-off - not the verdict

Recall and bare-arms improved; precision and stricter localisation performance declined.



**Conclusion:** M2 demonstrated a controlled performance trade-off, maintaining comparable mAP@50 while improving recall and bare-arms detection, with reductions in precision and mAP@50:95. As M2 was completed after the external challenge-testing phase, repeat external evaluation was outside the completed experimental scope; therefore, external-generalisation improvement is not inferred and remains a priority for future validation.

# A credible model is more than a metric

Traceability, licensing, evidence integrity and human oversight are part of the technical result.

### Reproducibility

Public GitHub structure; Colab-compatible notebook; README reproduction steps; Python 3.13.15; Ultralytics 8.4.156; Roboflow 1.5.0.

### Rights & licensing

Source dataset recorded as CC BY 4.0. Roboflow model pages record AGPL-3.0. External-image rights vary and should be respected.

### Responsible use

Decision support only. False negatives can miss genuine hazards; false positives can create unnecessary alerts. Human review remains necessary.

## Evidence integrity

**Known submission gaps are disclosed, not fabricated.**

The requested complete set of 10 validation-prediction screenshots was not preserved after credits were exhausted, and three defensible false-positive examples were not established. Genuine metrics, curves, annotation evidence and external challenge screenshots remain traceable.

Deployment boundary

**This proof-of-concept is intended as a decision-support tool and should not serve as an autonomous compliance determination or sole evidence of worker misconduct.**

# What the experiment established

Strong internal performance, visible external limitations, and a clear path to a more defensible system.

1

## Strong baseline

M1 reached 96.93% mAP50, but external tests showed that held-out metrics alone are insufficient.

2

## Targeted iteration

M2 improved recall and bare-arms performance, while sacrificing precision and mAP50-95.

3

## Next data priority

Add difficult Non-Helmet, bare-arms, Gloves, distant-worker, occluded and cropped examples.

4

## System priority

Strengthen worker-to-PPE association and explicit missing-PPE rule logic before operational use.

**Final position: promising decision-support prototype - not yet an autonomous safety authority.**